

ActInf ArtStream 001.1 ~ "Order and change in art", Sander Van de Cruys, Jacopo Frascaroli

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hello and welcome it is April 26th 2024 and we're in active inference art stream 1.1 starting a new art related series very appropriately with some great art discussions and presentations this is going to be order and change in art with opo frascaroli presenting and also with sander Vander croise and axel constant here for discussion so thank you all for joining looking forward to the presentation and discussion thank you Daniel thank you everyone for for being here uh it's great to be able to present uh to such a knowledgeable audience of active inference offici others and the work that we'll be presenting today uh is a work that Sander van uh car and myself did a few months ago and it is a work on a topic which I would say it's rather neglected or marginal so far in the active inference literature but uh as I will try to show it is a topic that uh is very promising it has a lot of interesting implications and indeed in recent years has been Gathering a lot of attention so let's jump straight into the topic and our starting point is uh the REM remarkable convergence of interest and results that has been emerging in the last few years between Scholars interested in the in mind and brain from a predictive processing or active inference perspectiv and Scholar interested in studying the Arts and Aesthetics from a variety of perspectives philosophy psychology Neuroscience but also increasingly the history of arts and artistic practice as well so this convergence of interests and results is fostering an ambitious research program that is proving beneficial for both sides because on the one hand predictive processing and active inferences are increasingly seen as uh Frameworks that can illuminate important aspects of our aesthetic encounters and our aesthetic experiences while on the other hand the Arts and Aesthetics seen from these predictive processing or active inference perspectives are increasingly seen as potential Windows into important aspects of our mental functioning so the talk will be exploring these two directions of research and most of what I will say uh is taken from these two papers uh one that uh sand van uh car Frist and myself wrote and that gives the title to this presentation and the other that uh I wrote together with s Helmut and El Helmut Le Anda brao and both

papers can be found in this uh recently published uh theme issue of philosophical transactions of the Royal Society B which is devoted entirely to these intersections between predictive processing and active inference on the one hand and the Arts and Aesthetics on the other hand so if you want a sort of a uh overview of the State of the art on the topic I definitely recommend checking out this theme issue uh the plan for the talk uh will be pretty straightforward as I said we will be exploring these two main directions of current research uh the first pertains to the ways in which we can illuminate aspects of aesthetic experience using the active inference and uh in this context we will be seeing what we mean by aesthetic experience uh we'll be introducing the notion of sense making and epistemic goals uh from this an active inference perspective and we will be talking about how these notions can clarify certain aspects of our engagement with the Arts and our aesthetic experiences in the second part of the talk I will be uh we'll be talking about ways in which on the other hand our mental functioning can be illuminated uh by the Arts and Aesthetics considered from this active inference or predictive processing perspective and we be uh examining a sort of analogy between what artists do and what cognitive scientists do when they study the mind and we will be clarifying how an examination of the Arts and Aesthetics can throw light on among other things the Dynamics of inference the Dynamics of affect and certain aspects of wellbeing in Psychopathology and in the end I will be presenting some conclusions and some directions for future research as well so let's start by saying something on how we can illuminate aesthetic experience from an active inference perspective now we should probably start by clarifying what's what is our explanatory Target here uh that is aesthetic experiences aesthetic experiences have been uh characterized by the philosophical and the psychological literature as experiences traditionally associated with the notion of beauty the perception of beauty in some sensory array and these are experiences that we undergo with varying degrees of intensity in the most varied circumstances of our life it could be that we are uh um looking at a wonderful Vista that we are examining uh and finding pleasure in the arrangement of furnitures in a living room but most Pro most uh characteristically aesthetic experiences are uh uh experiences that we undergo while engaging with art while listening to Great pieces of music while confronted with great uh uh work of of visual art while reading uh great novels and so on now there is a set of features that are recurrently attributed to aesthetic experience is and that's the feature these are the feature that we uh we are trying to explain from this active inference perspective uh first of all um aesthetic experiences are considered to be pleasurable these are experiences that uh are accompanied by the pleasant affective response normally this

doesn't mean that they are not uncomfortable or uh riddle with difficulty uh but in the end overall they tend to be pleasurable uh they are also self self- sustaining and with this by this I mean that uh uh they are experiences in which we seem to engage without any further purpose and in which we seem to engage uh in a continuous manner finding in this particular sensory array that we are examining uh noal things to examine and to be interested in and to uh care about as we go along and so they sustain our serious and attention uh throughout and they uh we seem to care about them with without any further utilitarian purpose in mind um aesthetic experiences are all often say to be transformative uh in the sense that when we when the painting is behind us the music is over when we put down the book or leave the theater we feel that we are not the same uh as before that we have kind changeed it and usually Chang changeed it for the best this uh this is a particularly strong trait associated with aesthetic experience that many uh many refer to and it can happen it can happen of course with various degrees of intensity it could be that you just felt like your world view has changed slightly or it could lead to massive Revelations about anything basically um aesthetic experiences are also said to be subjective in the sense that what is aesthetically appealing or pleasurable to me may not aesthetic be aesthetically appealing or pleasurable uh to you they are also said to be transient and evolving in the sense that they are characterized by an evolving uh um effective and cognitive profile that culminates normally with a pleasurable response but it's not that pleasurable response or judgment alone and they also evolving in the sense that what what I tend to prefer uh could change for me uh in time uh it's not just that what you prefer and what I prefer are different but also what I prefer In This Moment In Time can be different from what I prefer in another moment in time and also uh aesthetic experiences are tend tend to be seen as related in some way with uh uh feeling of freedom from existing constraints in our perceptual cognitive or motor process processes and a feeling of enhanced uh perception of our possibilities for Action we will see we will say more about that but these are more or less the features uh that we are we choose to concentrate on and they are more characteristic of of aesthetic experiences now uh the way in which we go about trying to explain these characteristics uh is uh by introducing these two Notions and trying the notion of sense making and epistemic ARs and trying to characterize them from an active infant perspectives so to understand them let's start with an example youve probably seen pictures like these before they are quite uh common in the psychological literature but perhaps you haven't seen this particular picture before if you haven't it would probably look like a random array of white and black Patches at the at the beginning but if you concentrate enough on on it after a

while you may start to not is that there is an animal in the picture most specifically a cow if you can see the cow here are some cues in the picture you just see the face of the cow uh these are the nostrils of the cow these are the eyes these are the ears of the cow now if you see the cow now you did something pretty remarkable already in the sense that you have arranged what seem to be like a random array of sensory stimulations into something meaningful and that's what we mean by sense making and we can like sort of break down this process uh from an active inference perspective by thinking about a sort of epistemic Arc involved in the discovery of structure in our sensory stimulation we start with uncertainty uncertainty about the potential causes of our sensory stimulations um these uncertainty in some cases in this case for example um triggers an expectation of the possibility to reduce such uncertainty and that's what curiosity looks like from an active inference perspective according to an increasing literature there is curiosity the feeling that that we can find structure in our sensorium that if we look uh uh for enough time we will be able to understand something is uh characterized from an active inference perspective uh as expected uncertainty reduction or expected Information Gain um which means the extent to which we uh we predicted we will be able to reduce uncertainty about the sensory the cause the causes of our sensory stimulations if we are curious that is in the in the cases in which uncertainty is such that we see uh the stimulus sort of yields a promise of making sense if we concentrate on it enough we will probably uh um try to uh to put in place epistemic actions there is actions uh that we we perform in order to reduce uncertainty and this could go from Cades as we uh examine the picture uh up to internet searches and every source of actions aimed at reducing the ambiguity or the uncertainty of our sensory stimulations and if we are lucky enough these epistemic actions leads us to uncertainty reduction and in the end we experience pleasure uh you may have noticed you will have notice that when you found the cow in the picture as with uh many of these Mooney images you are probably uh felt a positive affective reactions that psychologists normally call an AA moment or an AA experience that has both cognitive and effective components and and that accompanies the increase the the sudden uh clarity about the causes of your sensory stimulation so overall this experience generates a sort of Arc as we as we said that goes from uncertainty uh to pleasure and uh it is an arc this is import it is important to stress that this is an idolized experience because things could go wrong in many ways for example uh uncertainty in your sensory stimulation could be so high that you are not quite sure that you can reduce it and so you don't experience curiosity in the first place because you don't are not confident in the possibilities of reducing uncertainty in your stimulations but it

could also be the case that your epistemic actions do not lead to uncertainty reduction but to an increase in uncertainty instead after all it's a changeable world uh and even if we put in place actions to reduce uncertainty it could be it could well be that uh uncertainty stays the same or increase even and things could also go uh go wrong because um uncertainty in some cases is not reduced to an appreciable extent or is not reduced fast enough for us to experience this pleasurable effective response so as I said this epistemic ha is an idealized experience a way of capturing successful sense making we will say uh sense making that uptain in an OP way but things could could go wrong in many in many possible ways now if we think about why we experience pleasure while uh guessing the structure of our sensorium while settling on a specific hypothesis about the hidden causes of our sensory data well that uh has a pretty straightforward explanation from a active inference perspective because in this perspective as we know reducing uncertainty is pleasurable because it means maximizing model evidence there is maximizing the evidence for your own existence something which how and others have called self evidencing but it's important to stress and we stress in the paper uh that self-evidencing is not to be seeing a sort of passive process of confirming already pre-existing uh belief but it's more like an active process there is um it's a process of self transformation as well because in finding the structure of our sensorium in defining it in determining it we are also determining the structure of yourself as a model of that sensorium and so finding structure is always restructuring yourself in this way which is a very active process and that will be relevant in the in the in the rest of our discussion so reducing uncertainty is pleasurable and these grounds several theories of affect and wellbeing from a predictive processing or active inference perspective put some of the papers relevant papers there U they all more L stress that increasing prediction error or are normally effectively uh negatively balanced whereas decreases in prediction error are are normally seen as positive and uh these links well with certain uh assumptions and certain empirical results in the psychological literature and in literature in developmental robotic for example uh it is known for example that infants tend to concentrate on stimulations that are neither too unpredictable nor too predictable because these are the stimulation that maximize their their learning and their um the refining of their model of the world this is known as the goldilux effect and this sort of environment these spaces of the of these regions of the input space that afford optimal and certainty minimization are said sometimes called progress niches and we'll see the relevance for that for art in in a minute so overall there is a view that affect several aspect of our effective responses are related to the Dynamics of of uncertainty minimizations and that we experience positive effect uh uh in

relation to our successful attempts at uncertainty reduction and that this thing makes us concentrate uh to into regions towards such an optimal reduction in prediction minimization now let's try to apply all this to the case of art uh our contention in the paper that s has been developing for some years now um is that art provides us with good epistemic assets that is Art provides us with these uh idealized experiences of uncertainty minimization whereby we start with something which is ambiguous or not responding immediately responding to our uh predictions uh then we uh this thing has us to perform epistemic actions that are in the end successful and that leads to a pleasurable response if we for example take this painting by Monet we see this process going on at sort of high low level perceptual level which is um that the the ships and the water here appears so vividly because we uh probably puts in place the same thing the same process that we put in place in the case of the cow that is we have some ambiguous simulations something which not conform to the way in which normally water or or ships look to but if we guess hard enough we are able to see water and ships and then we get this AA moment uh this pleasurable effective response so what Monet is doing here at a low perceptual level is building such an idealized form of epistemic AR but of course that's another important point is that these epistemic ARS can go on at different levels of obstructions and here we can exploit the sort of hierarchical structure the active inference and and predictive processing insist on uh for example in certain paintings like this uh Yan Van painting here uh the matter is not guessing what figures are in the painting because these are represented realistically but the the point is to guess uh the meaning of the various attitudes the various characters in the picture or the various elements that that uh that uh that the the painter represent sense and their specific um position in the painting as well so these are higher level guesses uh which if successful lead to the same circle from uncertainty to pleasure that we saw on a perceptual level in the Monet painting and of course these guesses are not just happening at different levels uh but they are also different kinds for example uh in in in case for example of expressionist art uh the matter is not just to guess the meaning of the painting or the objects represented in it but to a larger extent to guess the intentions the state of Minds uh that the painter uh was experiencing or had while depicting uh uh himself in this case or any subjects uh uh and this is a form of guess ing of other mental uh states that is of course uh very important in the predictive processing literature and which uh Express expressionist that in particular leverage a lot but other guesses can have to do for example with the sort of movements that are implied or were put in place by the painter in order to produce the painting we have an example of the Famous Cuts By Lucha Fontana here but you can also think about

the thick brush Strokes by Van goog for example these allow these sort of artworks allow the viewer the viewers to uh make uh guesses about the actions that went into the production of the artwork itself so as you see the predictive processing active inference apparatus uh if leveraged to the full extent of its possibility can account for the pleasure that we get from different art Works in different styles in different moments in time as well now it's also important to to to understand then these guesses at various levels inform uh the guesses at the others level for example if you um if you have understood more or less the what figures are in the painting and you're making high level hypothesis about their meaning and and you and you make an high level hypothesis about something in particular you might be uh um pushed towards reexamining certain portions of the painting and noticing something on a perceptual level that you didn't notice before so often enough solving ambiguities or uncertainties at a lower level opens up problems on a higher level and solving problems on a higher level reopens problem on a perceptual level and this in successful Arts goes on and on and on and prolongs our engagement with the artwork itself importantly also artworks uh great artworks in particular are such that they keep us guessing uh they have a good supply of structural indeterminacy as uh Yuri lotman puts it there is they are uh ambiguous in many ways and such that if if you solve the ambiguities the first time that you see it uh you experience certain epistemic acts if you revisit it you may experience others epistemic act that you hadn't considered before and you can uh even uh experience more pleasure as a result uh but indeed uh great hards are seen as those sorts of inexhaustible stimulations that keep providing us with with uh important insights and that might be a way of characterizing how they do so in some if we take into account all all these ways in which uh art keep us guessing we will see that artworks are conceivable interpretable as hyper structural sensory streams um progress niches in their own right that allow us to keep uh improving our models of the world in very refined way and perhaps then the attraction that we have towards these stimulations is just part of a broader attraction that we talked before for regions of our input space that allows our optimal uh cognitive functioning now uh in this picture it is important to stress that aesthetic experiences are subjective in the sense that whether a specific sensory array will afford a pleasurable experience for a specific agent will depend on the generative model that it commands what is unpredictable or ambiguous or or uncertain for me can be perfectly predictable for you and so uh if that's the case it doesn't prompt the same epistemic AR that it prompts for me uh and we see these constantly uh in the development of the individual and Collective taste that often demands that we uh search for more ambiguous more unpredictable

simulations as we learn more about a specific art form in order for us to recover the sense of wonder and and and insight that certain particular simulation with the right kind of ambiguity gives us uh in this picture aesthetic experiences are also transient and evolving in the sense that they depend on the specific dynamics of uncertainty reduction and epistemic ARS involved um so for example for me this this Mony painting if seen many many times won't probably cause the same feeling of insight and pleasure that it caused the first time when I when I guessed the presence of the ships and the water in the painting much like uh seeing the cow for the second time the cow that you saw before uh doesn't prompt the same aha experience to the same extent because now you are more likely to directly interpret it as a cow in an automatic way and so you see that uh aesthetic experiences are even with the same stimulus can be very transient and evolving and in a sense uh we are always trying with aesthetic stimulations to un to recover the sort of Ingenuity and and uh that we had while seeing the picture or the stimulation for the first time and art is often characterized as providing us occasions to see things as for the first time and that could be a way in which it does so now the last thing I I will say about um aesthetic experience in from this um active inference perspective is uh that as we saw aesthetic experiences are often characterized as having something to do with freedom of enhanced or enhanced uh sense of agency for example K talks about the free play of cognitive faculties that will be involved in the perception of a of a beautiful object uh Shila even goes as far as saying that it is only through beauty that make makes his way to Freedom so there is an ancient philosophical view that links aesthetic appreciation experiences uh to freedom from existing constraints and an perception of our possibilities for Action how do we explain that from an active inference perspectives well there are several ways of accounting for that aspect of our aesthetic experience one is that if the the the picture that we gave so far is right uh aesthetic experience are occasion of enhanced self structuring in the sense that they are moment where we determine the structure of our sensorium at a faster rate than usual and in doing so we determine our own structure in a faster rate than usual in a sense we keep building ourselves we keep uh changing uh we free oursel from existing uh ways of perceiving things and we turn our minds into moving targets uh while we engage with these stimulations so uh these probably captur something about the notion of autonomy that can't had in mind also when speaking about arts and that is intriguingly finding new application in cognitive science as well and activist for example insist on this notion of autonomy as well which is giving yourself your own Rule and perhaps in aesthetic experiences we are doing that to an enhanced

degree because we are determining the structure of the sensorium and the structure of ourselves as models of that sensorium in a faster rate at a faster rate than usual but another way of thinking about how aesthetic experiences can boost our sense of Freedom an agency is that these experiences are normally uh conducted uh or encountered in the absence of pressing needs that is we have no pressing needs that push us to organize our sensorium in precise ways and we are free to explore many hypothesis about the underlying structure of our stimulations and this is probably what people in Aesthetics uh uh talk about when they talk about the aesthetic disinterestedness there is that specific state of mind that one has to have When approaching something aesthetically which is an a state of mind a mind of Disconnect with practical interest or interest of of other kind and this free up space to explore other possibilities of perceptual and cognitive organization another thing that one can take from the active inference uh framework uh in order to explain this relationship between aesthetic uh experience and freedom is that uh perhaps as we speculate in the paper uh when we experience a sudden reduction in prediction in in uncertainty these in a c in in a certain sense boost our confidence in the ability to reduce uncertainty in the future there are interesting uh Studies by you and colleagues for example our Barbarian colleagues that show in this case for example that um even an AA experience of the kind that you experience with a cow can um encourage risky decision making in in uh in in a in a in a subsequent task and this study by barbar and colleagues shows that the stimulate that we tend to uh like uh are also accompanied or Foster uh more curiosity so in a sense one can can see successful aesthetic experiences as something that allows us to um broaden that epistemic H That We examined before uh and because we have more confidence that we will be able if we engage with epistemic actions for subsequent for for uh for the necessary time to reduce uncertainty and we all face experiences in which these epistemic ARS are very short uh in the in the paper we do we make the example of internet memes which afford this very short epistemic Act of of uncertainty and Discovery but art is normally seen as boosting our confidence in our ability to endurance and and and uh uncertainty something which is sometimes uh referred as negative capability and and so that's perhaps enough an interesting effect that art could have if examined from this uh active inference perspective now summing up this part of the of the talk we have seen that the active inference of predictive protic apparatus helps us explain why aesthetic experiences are pleasurable namely because they are instances of successful self- evidencing uh because they are sustain why they are sustain self sustaining because they are made of uh they may they are made possible by sensory streams that fuel our curiosity and keep us guessing they are

transformative because they are instances of announced self structuring they are subjective because they depend on the agent specific generative model they are transient and evolving because they depend on the specific dynamics of uncertainty reduction and they are linked to free freedom and announced agency because they are instances of announced self structuring again conducting in the absence of pressing needs and that boost our confidence that as uncertainty minim minimizes now for the second part of the talk which is slightly shorter uh we will examining ways in which our mental functioning can be illuminated by the Arts and Aesthetics if we consider them from this predictive processing perspective now it's important to see that in this framework artists uh are seen as experts in cognition and in inference in particular in the sense that they know what cues ought to be presenting to our sensory organs for us to formulate certain evolving hypothesis about what we are sensing um a painter for example knows what sensory array to produce in order for you to guess that there is a certain object in the painting but not just that in order for you to guess it with a certain profile of uncertainty minimization and artist have a sort of implicit largely implicit knowledge of these dynamics of uncertainty minimization for example they can feel the tensions the visual tensions of an imbalanced picture they can tell what notes or what word are we expect uh uh in a in a in a sentence or in a melody and they can craft cinematic flows with the right kind of abruptness or uh invisibility in in the editing for example so every artist has in this framework sort of large a large body of implicit knowledge about the Dynamics of inference and indeed art is probably a place where these dynamics of inference and their cognitive and affective correlates become particularly visible in the sense that all the what artists are doing is to manipulate these Dynamics in order uh to produce on us cognitive and effective reactions so therefore the thesis is that artists and aestheticians to the extent that they have access to this large body of implicit knowledge might therefore contribute to our understanding of a mod things the Dynamics of inference the Dynamics of affect and certain aspect of wellbeing and psychopathology so let's let's examine these three things in turn well as we said probably artist assess this implicit understanding of our inferential Dynamics for example if the Moz here crafts a Melody he knows whether certain note is to be expected given the preceding context or whether to introduce another note which is not expected given the predictive context in order to generate certain effects in fact some people characterize uh music theory as the most developed psychology folk psychology currently extent exactly to capture this fact that in within artistic practice there is embedded a large body of knowledge about our Dynamic the Dynamics of our inferences uh and the same goes for a poet for example this is not a particularly poetic

example but for a poet that chooses the carefully the words the words of of his poem in order to generate certain profiles of minimization so one can say that artists are subtle manipulations of the inference uh we make and the timing with which we make them and this knowledge can indeed be exploited to link the personal the subpersonal level uh when we examine experience for example and this is an interesting point many of these violation of our melodic or semantic expectations have neural markers that can be register with electroencephalography as you see here in the picture and so by linking these uh neural correlates with the the phenomenological variations that they are accompanied uh by which artist and aestheticians examine perhaps we can have a a means to contribute the literature on the Neural correlates of Consciousness or to that um scientific Enterprise that goes under by the name of neurophenomenology that does exactly this operation of connecting sub personal and person level experience so ad could give us access to ways of doing that now as we said from a predictive processing or active influence perspective cognition and perception are always accompanied by affect uh because they are about uh our existence as viable models of the world so in mastering the Dynamics of inference uh artists are probably also mastering the the Dynamics of affect and particularly the Dynamics of those uh emotions or those affective states that are sometimes called epistemic or metacognitive emotions things like surprise confusion curiosity uncertainty bottom Insight this fluency and so on many of these things interestingly are are starting to be examined from a predictive processing perspective and and again art could be a way in which we bring uh the expertise of artists and aestheticians about these these states and the way they are produced uh to bear on Research in cognitive science that examine the same thing and this could also exam inform debates in philosophy and cognitive science about things like affective engineering affective scaffolding and extended effectivity there is ways of conceptualizing the many ways in which we we um modify our effective states by modifying things in the environment and we manage our effective states by uh managing the sort of simulation that we encounter art seems to be a particularly interesting case of effective scaffolding or extended effectivity and it is interesting to to to to see that indeed art is starting to be examined from that perspective in some recent work as well and finally as for well-being and copathology well we saw that aesthetic experiences if we are right have a lot to do with optimal cognitive functioning in the sense that they afford those optimal um experiences that we characterized as epistemic CS uh they are about optimal information seeking about optimal development of our mental models and so to the extent the that artists and aestheticians have access to these Dynamics perhaps we can gather for them indication on how to design experiences that favor

involvement motivation learning and Discovery and this is indeed an increasing stream of research uh that goes sometimes under the name of art therapy um and there are many studies examining the effects of the exposure to Artistic stimuli on several cognitive faculties and abilities that seems to go in that direction in the direction of suggesting that artist has some sort of knowledge on how to arrange sensory streams that boost these things Discovery and finally uh perhaps if aesthetic experiences have a lot to do with optimal cognitive functioning they can perhaps shed light on on on forms of uh path pathological cognitive functioning uh or epistemic Behavior Uh um because indeed one thing that active inference allows us to do is to conceptualize many pathologies of our information seeking Collective information seeking like Echo Chambers conspiracy theories and so on and many psychopathologies like schizophrenia delusion autism anxiety and mood disorders like anxiety and depression is having to do something with suboptimal ways of engaging with Environmental uncertainty so if art can tell us something about optimal cognitive function perhaps it can tell us something about deviations from the this optimal cognitive functioning which it is important to stress are often put in place in order to recover that viability in the end and and on the other hand studies about uh uh these pathological conditions could also inform the production uh the study on the of the production of Art and and aesthetic experiences um because as you know there is a long tradition of uh of studies linking creativity in in act with psychopathological conditions for example okay so let's draw some conclusion now now and some direction for future research uh by just reverse rehearsing what we said so far uh so we have seen that there is a remarkable convergence of interest and results between scholar working within the inference framework and Scholars studying the Arts and Aesthetics this convergence is proving beneficial for both sides because on the one hand predictive processing and an active inference are Illuminating aspects of our aesthetic experiences things like why they are pleasurable why they are self- sustaining transformative subjective transient and linked to freedom and anoun agency uh apologies for the the misspell and on the other hand art and Aesthetics are beginning to be seen as Windows into mental functioning and revealing something about the Dynamics of inference the Dynamics of affet and well-being and psychopathology now to point out to direction for future research uh in Aesthetics in particular which is a big field that has uh contributors from philosophy psychology uh neuroscience and so on uh the next thing to to study probably from this predictive processing or active inference perspective or the next thing that are needed are a more empirical testing of the predictive processing hypothesis which are sometimes very difficult to test in in the in the in the

introduction to the theme issue we examine some of these difficulties which have to do with the mainly with the way of uh um Gathering measures of uncertainty and uncertainty minimization for example uh so more empirical testing is needed but also more insights into the neural underpinnings of these Dynamics uh which are quite often linked to the limbic systems and and the relationship between uh the way in which we conceptualize dopamine and other hormones from a predictive processing perspective but the neural underpinnings of this epistemic as these aha moments are still all to be clarified uh also it is important to understand how these predictive processing perspective that we are advancing connects with uh links to other Frameworks in empirical Aesthetics which is a big field with a long history H and this is particularly important if you want to to discern uh the specific predictions that have to do with the predictive processing framework from predictions uh that are made in other Frameworks as well and and important also expansion expansion of the story would be its extension to not just exteroceptive stimulation uh which have been the subject of our attention in the presentation and in our papers mostly uh that is stimulation that we get from the environment but also proprioceptive and interoceptive uh stimulations that have also their own inferential Dynamics and they could open possibilities is for example to conceptualize also things like sports performative Arts the playing of instruments for example or skillful motor Behavior as a way of as as as a form of aesthetic experience itself and another relevant expansion of the story which hasn't been covered so far a lot is taking the artist perspective because we we talked a lot about aesthetic experience there is experience of subjects exposed to aesthetic stimulations but we haven't talked much about uh the way in which these stimulation are produced that is the the creative the creative process that uh it's put in place by artists in order to generate these carefully curated sensory streams and there is a very interesting beginning of that uh of that exploration in the paper by by by by Axel Constant and colleagues in the theme issue which I recommend uh I refer you for more on creativity and and active influence more broadly um the expansions of the view that we that I am presenting for cognitive science have to do as we said already with uh studying the Dynamics of inference after and what being in Psychopathology which are all directions that are very much still to be explored but also there are many other phenomena that are normally examined under an aesthetic perspective but they that have also been uh given accounts from a predictive processing perspective things like learning attention motivation and conscious experience that are very much still to be examined uh from this perspective and the need and that will probably profit from the encounter between Aesthetics and and art on the one hand and active

inference on the other okay I conclude by just uh reminding you again of the theme issue on these topics and thanking once more my uh collaborators uh first of all sand van C uh which is also here will be able to um to to to stay for the Q&A and answer your concerns as well car friston Helmut Le and atico thank you very much for your attention I look forward to an interesting discussion thank you thanks for the presentation awesome thank you all right well lots of places to jump in perhaps sander first and then Axel if you would like to make any initial remarks or just give any context or what you are left with from the presentation uh well it was a great overview of uh of the work we've done together and um I have little to add immediately um but but yeah I need I'm I'm interested in like how can we emperically test this and how can we quantify uncertainty in a way that um we can link it to people's experiences uh and and in music This is this is happening a bit like people are using um models that output measures of while uh while processing progressions of ports for example uh this the work of of Chun um and and there it's found that these models the surprise measures and the uncertainty measures that are derived from from these models they uh predict the emotions that that individuals are are experien when when listening to this music um so that so that's an interesting um an interesting step and and next step I guess I guess would be to look at it at a more more personalized way in in in which we for example have Jazz lovers and we we we train this this computational model on jazz music and and we see whether the surpris the uncertainty that that kind of model output that that it it relinks better to the emotions of the Jazz uh lovers uh than to General uh listeners so so then it's really tuning the uncertainty to to the to the individual or or the group in this in this case so that that's for me a very promising and interesting um route to take next uh but but in general I think I'm I'm happy with with the story that jao told so I I want to hear your reactions great Axel yeah well thanks ja for for talk it was um really interesting um so yeah I guess I have two large sets of questions um uh one of which you already sort of um highlighted which is about creativity right relation between the creativity as um the the you know creativity as that which as an attribute of the artist um product or process and that's how you sort of operationalize it uh in Psychology uh sometimes uh you will have the creativity of the thing that you sort of uh produce that is a creative product or a creative process and that is my my experience as someone who who produced that thing that is creative you know that can also be creative in a sense right um and um the first well the set of questions that have to do with creativity for me it's especially this interesting observation that um the creativity of the process doesn't seem to be predictive of the creativity of the product right uh because product creativity you have like two criteria the

aptness and the novelty right and the aptness has to do with uh let's say I want to create a um a stand for a laptop it can be a creative laptop stand uh but uh for that um stand to be uh creative uh as a product quote quote um it also has to work also has to hold my laptop right so that's the appness Criterion has to be useful and has to work for what um it designed it's designed to to do right and then you have the novelty which is a statistical Criterion which is from the point of view of all the kind of laptop stands that have been produced uh which of those are considered you know most original if you will from the point of view of the people would assess all the population of laptop stands right um and again that doesn't seem to track with um my creativity the creativity of the process of me producing laptop stands right so I can I can think or feel like I'm producing a pretty boring laptop stand and it turns out to be a very creative product creative laptop stand right so I would be interested in uh hearing you on the the sort of um you know not a contradiction but something that is slightly unexpected right um so that's the first set of questions concerning creativity it revolves around this um finding interesting finding and then the other set of questions um has to do with and I think this is a more General problem in the predictive processing literature uh and aesthetic aesthetic experience um we often say that for you it's the concept of the aesthetic Arc right and hearing hearing you it it um it sounds as though um traveling on the arc if you will so reducing my uncertainty would in and of itself be pleasurable right and I'm not very clear on why that would be the case I I I would like that to be the case because it it sound very intuitive right when I reduce my uncertainty it's like it's a you know it's it's it's good it's I like that it works well right but really for that to be the case uh there would be uh we would need kind of a story about how how um computational neuro uh computational physiology according to active inference right maps on to the actual physi neurophysiology of pleasure right so I'm not sure we have that story or maybe we do but I I haven't seen it right or the the the alternative would be something like um once I'm at the end of that sort travel on that Arc right I ached at an outcome which will be pleasurable or reward and that works for me right but the art in and of itself because it's sort of it's a process of minimizing uncertainty it would be there for pleasurable I I I would need more um explanation of why that would be the case right so I guess yeah those are the two sets of questions thank you very excellent questions uh I'll probably I'll start if you don't mind asand and then you'll probably add more especially perhaps about the the second question as well but in any case yes so um creativity process creativity versus uh product creativity uh I think the discrepancy that you point out could be characterized as thinking again about the fact that people have different gener command

different generative models and have uh for example this could lead to different views about what's a great way of arranging a sensory array in order to produce uh certain results uh an artist could feel that especially if it has practiced similar procedures all all over all very frequently that the solution at which he he arrives or she arrives at is boring or uninteresting um and the audience may feel very different uh for the fact that they the generative model is different they don't expect a particular solution uh perhaps uh even it's often the case that this the solution has been arrived at serendipitously uh the the designer hasn't accounted for certain uses or certain way of reading the stimulation that people find in the stimulation themselves because they command different generative models and and so perhaps there are ways of accounting for this discrepancy between product creativity and and process creativity but it's very interesting I I I've always found interesting the way in which product creativity is described as you said as being as having to do with OP both openness and Novelty because it seems that these two things capture two aspects of the story that we have been telling uh novelty of course captures the the sort of uncertainty uh unpredictability of a certain sensory ARR and the way which you should interact with it whereas openness captures the sort of the the end of the the parable the when when you get the insight and you get how all the the the pieces should should fit together and and you need to have both in order to have an aesthetic experience in our account and in order to be creative so perhaps there is the nice there is the nice link here between the aesthetic experience uh of a viewer and the creative process the generated that experience right so it would be like I had this laptop stand that I look at it and I'm like what's that and I I start try to use that thing and then I figure out that it is a laptop stand and that whole process right um you know once I figured out that it is a laptop stand and it works well as a laptop stand right the aptness Criterion is met uh but I had to to get to that point I had to go through a process of uncertainty minimization because it was such an original in novel exactly because you notice openness more in a sense if it comes into a Context in which you need to Discover it uh otherwise you are just acting automatically on an object that you already know how to use you can perceive the creativity or the the object even a MAG is creative in a sense because it affords a certain uh way of using it but you need to recover in a sense that creativity that past generations of makers have put into that object which is is not visible to you anymore in a sense so again you have a a sort of relationship between U creative processes and which looks create look or do not look creative according to how habituated we are to them and the aesthetic experience as we conceptualize it which has to do with uh stimulate that are novel enough for us to notice them and but not so novel that we we

are not able to use them anymore just I would just add to that because that's very important to to see like the aesthetic experience also as a creative act in a sense that this blurring of the of the line between the Creator and the perceiver uh is is is important in this view that that every aesthetic experience is in a sense because of this discovery aspect and and a creative act as well um and and so the appness is very much dependent on yeah the models of of the of the viewer are the models of the Creator but I mean that that there is discrepancy outside and inside that's like a a logical thing to me um and of course as much as the models overlap between these the audience and and the artist and uh then there will be agreed upon creativity um strength as it were of of the the object but but it might come across as very creative for one person but not not at all for another person so yeah yeah and it it you know you often hear stuff like especially for things like Contemporary Art right uh where people will say I don't see the point I don't see why that's interesting right uh whereas for another person who might be some somewhat of an expert in contemporary art that person would see how interesting a novel you know as a product that piece of work is right so it it sounds like um even though you you can explain why a creative process does not necessarily lead lead to a creative product right but there must be some kind of you know there has to be some kind of um you know you know tracking or correlation between the two models um the model of the perceiver and the model of the artist that has produced right the product or the piece of work because otherwise you fall into those situation where people they just don't understand what's going on right and then there's no aesthetic pleasure or aesthetic experience right so then the question is you know is there a sweet spot of um kind of mismatch between because that would that would be exactly what accounts for the possibility of an aesthetic art right there must be some kind of mismatch between um the model the artist that produce the product and the model the perceiver right that uh uh that you know perceives the product and because if there's no mismatch it would just be like a clear communication like a stop sign right uh the model of uh the uh the government that produced the stop sign matches perfectly your model of what a stop sign is and so there isn't really a sort of an aesthetic Arc to go through to understand you know the meaning of the stop sign right so the question becomes what is the The Sweet Spot basically in terms of mismatch between uh the model of the artist and the model of the perceiver right and then maybe you can look at art history and what what typically is the sort of educational background of great artists what kind of a typical experience they had in their life and what kind of model that could have led to and how mismatched that type of model right is with the model of population at a certain time in history right so uh yeah yeah and and

it's I mean that it's also like the culturally dominant model that is important there and and you see that artists will play around and will hit and miss in these kind of things I mean they will experiment a lot uh and and be uh be be uh early on their on their own times like like they they they might um propose like some some forms of uncertainty that culture cannot deal with at that time and a lot of the those things we we just don't know about of course because because they're not filtered by history it's only afterwards that we we realize like these Dynamics we're playing we're playing and and he found the The Sweet Spot but but uh yeah it's it's not in advance I guess or or very hard in advance to uh to [Music] predict yeah I would add that there is a whole field like in aesthetic theory on in the theory of art which is call like the Aesthetics of reception which whose role is exactly to study the sort of system of expectations that a certain audience has at a certain moment in time and the way in which the new artistic product interacts with these expectations uh by uh violating them um to a larger or lesser extent and and and so there is already a tradition of studies that uh understood that if you V at things too much the reception is not going to be good if you violate them uh too little it's not going to be interesting so this the sort of Sweet Spot talk that you that you adopted is very much used in that literature as well and of course this active inference story has the merits of uh connecting those early intuitions with this neurocomputational language and and all the literature on all the other cognitive pH phoma that I explained within the same framework I'd say and as for the second also very interesting uh um question uh if I understood you well you gave two Alternatives either you have pleasure within the all uh along the all ha that we have described or you have pleasure in the end uh right or to explain the relation between pleasure and E estimate art would it would be to say well there's kind of a there's a process theory of active inference right so it speaks also to actual neurophysiology whether that theory holds that's a you know that's an open question but you know for the claim to work that computational uh uh neurophysiology would have to match the neurophysiology of pleasure right I don't know if that's the case or the alternative is well it's not about the arc it's about the end product and the arc is what you need to get to the end product right so yeah yeah it's very interesting I agree uh and that's one of the problem that I was hinting at and in the end that there is no clear story to my knowledge but my knowledge is very limited because I have a philosophic I'm philosopher by training um about how to map these things into the what is known about the neurophysiology of pleasure the um the limic system there there is some some work in that direction uh we we mentioned some of it in in the in the paper that has to do with dopam dopaminergic discharge during the moment I guess so in the end that would be in

the end of the process uh um but conceptually at least what sustains the attention of the viewer throughout the process is first of all curiosity is not pleasure or not pleasure in a traditional sense but the expectation that things are going to uh make sense in the end if we take enough effort to make to to observe them and and the other half is the the pleasure half I would say in which uncertainty if you are lucky gets reduced and gives you this output but yeah I'm curious to to to to to see what what sand thinks about the question as well yeah I'm not I think uh if you talk about neurophysiology of course active inference has a a whole other take on the dopamin arch system and and all all these reward systems but but but um in a sense I I think the the positive effect is just um like you talked about this confidence boost after restructuring after sudden restructuring I think the the the positive effect is just this feeling of the confidence and the opening up of of Alternatives which probably has to do with with kind of um a a Precision that is um yeah in a diffuse way um projecting back to to all the things that you you are doing but but in what way specifically that that's that's very unclear to me like but it seems to be that dopamine dopamine might be well placed or these systems might be well placed to have a diffuse effect on further processing in a kind of uh yeah uh energizing sense but but but I'm not thinking about pleasure as the traditional reward sense uh I don't I don't think that's very useful in in a in a netive inference uh framing um yeah because yeah it it doesn't really explain much if you yeah if if you just point to a reward center that is then activated and so so so I I I try to see it as more as as a kind of network effect and a kind of um yeah as you discussed this sub subsequent effects of of an AA that then um boost your your your autonomy by by having these effects on on Precision uh but yeah that's all very speculative still I think yeah um so so two points Jack you said um having a sustained attention right so maybe in what we call aesthetic experience there there needs to be sufficient sufficiently sustained attention for at Le for long enough to get to that sort of rewarding but not rewarding point but that end point uh Center that you're talking about right so maybe that's the the you know the the ep the epistemic aspect of that Arc um it its function would be to maintain the say attention for long enough and now the question is you know what what what does that afford to to that experience is and then an answer could be something like uh there's research on how in those models the volatility in your beliefs about the structure of the world that is your transition Matrix that relate to stuff like anxiety right from the point of view of neurophysiology right so maybe that what we call aesthetic pleasure is just the release uh or the you know the reduction of a feeling of anxiety and uncertainty right and the a moment it's not like a um a of knowledge it's a a of relief right so it's U I don't know maybe maybe that's

how you can conciliate the but at any rate you would have to make um or people who make that claim that the arc is pleasurable because reducing uncertainty is pleasurable I again I'm not and I know Senator that's not that's not your claim but I feel like that in the literature we sometimes read that minimizing the uncertainty per se is pleasurable I doubt that that's the case maybe has that is you know reduce anxiety makes you in a you know you know in in a in a state that you're poised to you know having the sort of pleasurable endpoint or something like that yeah that's an important point that that that maybe wasn't emphasized enough but indeed like if if if active inference is true then we're constantly reducing uncertainty by actions by walking and so it cannot be merely that it has to do with some kind of unexpected rate of of of uncertainty reduction so so they have have to be that's why artists work with with obstacles and and and and in creating uncertainty to to yeah to try to have us this experience um but always in a in a like a fallible way of course but to try to give us the experience that that we s ly resolve the uncertainty in a way that is faster than than we expected maybe when engaging with it first yeah but that that's just that just sort of pushes the problem upward right it's not this first level of uncertainty and the sort of it's the uncertainty over the uncertainty but you know if you say well that's you know the pleasure comes from reducing the uncertainty over the uncertainty you haven't solved the problem right you just sort of further push it upward in the hierarchy basically um I guess the problem is that we don't have a good story of why um reducing uncertainty per se would be tied to like pleasurable experience right um and and what what what what would what would you find like a a um a kind of satisfying explanation because for me it's not so much that like it it it seems like um this it is more puzzling to me to start from a rewards perspective where you people get rewarded by some things uh Suddenly by having some activity and reward structures um the the advantage of this story is that that it's not it's not about the reward but it's about this engagement with the environment that that is uh positively or negatively balanced depending on how how well we are doing in in this uh structuring or predicting it uh so so so yeah I guess I guess maybe your question is then like what is still the function of the positive experience uh do we need it at all if it's um can it be just an epic phenomenon or something uh and that's a fair questions question but but but uh yeah I think that we have to look at the the consequences of the positive effective experience and and and not so much yeah at I don't know um I does that does that make sense to you to that kind of uh or what's your question about the the the the phenomenology of of of effect or right I mean I I would have I could respond something but I maybe then there there are people that have uh questions in the chat or do you want to I think

there's like 14 minutes left or so so if you want to I think this points to some of the the very complex and interwoven aspects of the art viewing experience and separating out like primary sensory reductions of uncertainty then there's art pieces that leave us with questions so then okay well then that is increasing our Precision on us being the kind of cultural thing that's left with a question after the Shakespeare play those start getting into very stacked and and socially enacted kinds of uncertainties and actions and Pleasures that may be very distal from the laptop stand it could be like like oh it's it's giving me a feeling like there was care in the production of the artifact or I know the person who made it so then the eye that the parent brings to the child's art piece is different than the art reviewer for an show and so it's um almost like an invitation to explore those alternative hypotheses now within a space where we can at least jux depose them and recombine them because there's so many media specific and and experiential elements that can't necessarily be separated out very cleanly because we're talking about the human reception of art not a um insect's reception of two solid colors something that that might be more isolatable to elements like ambiguity or luminance or more basic properties of the stimuli and then I think the one the example you showed with a torn canvas how that's a trace left by a process um and then the stop sign too I mean in a way modern art has kind of broken Bren that that fourth wall with the urinal and with the with the blank squares and all these kinds of things that that complexify just the day-to-day engagement and for some people there it's a bizarre um unnecessary complexification for other people it's kind of a re-enchantment with regular objects and it just shows how even an object that somebody did not intend to produce as art could still trigger a deeply aesthetic experience and so it's almost like more questions than answers perhaps nice yeah I mean yes one of the advantages of the story we have been presenting I think is that active inference offers is possibility to exploit the hierarchical organization of our predictions uh of course as predictions becomes more abstract and are also more difficult to to test and to to verify and we enters into a a lot of very difficult to to pinpoint predictions uh which are cultural in nature uh uh that that have enough effect of our appreciation of of a particular artwork but we are not quite able to pin point but the idea is that with for example with abstr art a lot of the way in which they function as aesthetic stimulations is by playing with our very higher level expectations about for example how an artwork would look like according to a specific art world according to a specific community of art uh consumers and if Duan for example could make art by by signing a neuronal it is because he was playing with these very high level expectations about what art is supposed to look like it's like a challenge it could be

conceptualized among other things as a challenge to see that P specific object as an artwork in the same way in which I I suggested that a particular sensory array could be seen as a cow so if you manage to do so to see that specific object an artwork you get that insight which is similar to that sort of low level insight that you got with the cow but on a higher level and um yes so one could exploit this hierarchical story in order to account for different things among among them uh abstract art as well there is a very interesting piece on abstract art in the theme issue by a heart historian Ladislav Kesner uh so that would be another thing to look uh into for example yeah yeah one one comment from the live chat Holly wrote when thinking about abstract painting the viewer might find pleasure in compositional elements like variety contrast repetition unity and balance in addition to the interpretation and gesture yeah sure I mean uh of course there are also these uh these uh things about the arrangement of of elements in in the canvas the way in which they they create a balanced or unbalanced design how we explain the the the intentions of the painter to position these elements in this particular way of course if we know more about this particular poetic of that artist if you for example it becomes vital with certain artists like Mondrian or Kandinsky to know also about the theoretical writings in order to make sense of why they are using certain colors and certain shapes in certain forms so it's also a case that with abstr art what is sometimes called the the the broader context the epitext all the things the information are around the artwork gain more importance because there are ways in which artists set expectations on us not by things that they put in the artwork themselves but by presenting them within a certain context of theories and so on so the a lot of predictions are installed by artist in in abstract art in this epitext more than in in the work itself awesome if I quick can quickly add on because this is a point I wanted to raise is that um now we're talking about aesthetic experience um I think you can appreciate our outside of aesthetic experience as you said ja so you can appreciate let's say an artwork as a art historical object and then you have this sort of a intellectual pleasure with it uh and even with you when you look at uh let's say conceptual Arts right uh there's this uh well um what will happen often times is that they will use this device called a citation right whereas in the piece of the artwork there will be a citation to earlier work and you can appreciate the citation if you know the art history right so um but in that case you're kind of outside I well and that's the beauty of active inference are you really outside of the domain of aesthetic experience maybe you can make claim that you're you're within the same sort of regime of like computational processes and ctive processes sure but um let's say in the common sense of aesthetic experience then your experience you know relies much less on the sort of um rolling

uh you know sensory processing Dynamics right that you would have in something like I don't know um you were mentioning um abstract and uh you know Kandinsky and where you have a lot of colors you have difficulties interpreting the the the the figure and so on and so forth right so um maybe it would be useful to talk of different aesthetic experience and for different objects and different ways of appreciating those objects and that might help clarifying right that maybe different um uh you know trajectories on the epistemic arc right and so on and so forth so all the kind of problems that we discussed today maybe that be a good starting point with us the tax on to me like that yeah I mean I don't know whether s agrees but I think we are all always in the same sort of aesthetic domain there are no differences if not in this in sort of abstractions in here the the nature of predictions that you put uh into play in the different cases between appreciating something for its lowlevel features and the way in which they are combinable and having this very abstract high level knowledge of the art scene that informs your guesses uh and your appreciation of artworks also because these processes and these guesses inform one another so you need to have sort of plastic view of where the bounding is or or not having boundaries at all between these two things uh because for example if you know for example the title of Picasso bullsh Head the thing sort of sculptur for by pieces of a bicycle if you know the title that's sort of a high level information about the artwork uh we are also inclined to concentrate on certain sensory features of the object instead of others and to make some lowlevel inferences instead of others and if you make those inference perhaps you formulate new high level hypothesis and these two things continuously inform one another uh and that's the same process to me as long as you have these epistemic ARS that leads you towards some sort of discovery which could be high very high level or low level I mean to me it's one of the advantages of the framework I don't know whether sand agrees or not yeah completely yeah I I would even yeah make it more broadly I mean the the even in games uh you have these epistemic arcs and and just in everyday life they they might I mean we we call them the like a minimal unit of sense making and and it is it is kind of that so it's not really necessarily attached to Art um but it is a kind of uh pleasure that is not derived from like uh food or or uh sex or the the the kind of typical rewards that people use in Psychology um um so so so and and that's the advantage of course of active inference that it that it makes room for this kind of like I would call it a pleasure of obstacle clearing in in in a general sense to towards what you want instead of getting what you want it's it's like we want to problem solve in order to to get where we want and I think it makes sense for for evolution to to reward like this this obstacle clearing and this this overcoming challenges which is of of course

uncertainty so of overcoming uncertainty is is the reward as such um and it makes sense I think for evolution to to to organize us as as such and maybe as as humans more than than other animals but you see them also like I I give the example of my dog that that often picks the the largest kind of unwieldy stick to to play with and I see that happen and I I I think it's probably because it gives him the most Challenge and if it gives him an obstacle to overcome and if I don't want to play with him uh he has to find his own obstacles and he has to find ways to reduce uncertainty in display uh that that that he generates himself and and that's art as well of course um so so yeah that's for me the the the strength of the the approach in inference yeah if I can add very brief remark um that that's in to some extent that's going back to the sort of modern origin of Aesthetics to Bard and who used to think of Aesthetics as the science of sensus sensus cognition he said in general um so of course the difference between everyday experience and aesthetic experience will be that aesthetic experiences or aesthetic objects present ourselves with these more idealized form of uncertainty minimization that we tended to characterize with The epistemic Arc and I I showed how things could go wrong in many ways ideally in art when it when when art works we get this epistemic us in a very uh purified and idealized form and that's what why these experiences are more pleasurable but indeed the the component the the inferential component of that experience is something that we find in experience more broadly because as as all experience is finding structure in our sensorium but not all experience is as successful in in finding structure in our sensorium as aesthetic experiences are awesome well in closing if you have any last thoughts or any short ideas or visions for how the art stream series might continue on its own epistemic path so let's do yeah jacob a first then sander and axel okay thank you um yes I just hope that this were this was very non-technical overview of the field I just hope that it got people interested in active inference in interested in this work on Art and Aesthetics in particular because uh I think it holds lots of promises interesting applications and I look forward to the the rest of the this art stream that this U first uh encounter inaugurates and uh an obvious solution for me would be to contact the authors of the theme issue that I talked about uh but there are also several people that are trying to test this thing experimentally as well in psychological or neuroscientific labs and if you need any help in contacting them or or in uh finding out people uh for for for the art stream I'm I'm here of course thank you CER yeah I second that I I think it would be nice to uh to to try to work towards um more empirical ideas and to see what people have have done and how how we can build on that um because yeah that in in the end that that's that's the promise of active inference if it if it doesn't allow us to get a better handle on like uh uncertainty and

quantifying uncertainty relating it to the phenomenology during art perception then there is I mean then people might say it's just a rehearsal of some class ideas about expectation violation in art and so it's really like the the kind of and also of course the kind of AI models that we now have to to to to uh use as a proxy for uncertainty and surprise and to see whether that captures something in Human Experience uh if we if we start to use text or music as I said uh so so yeah that's really the the advantage of of the approach and and the promise um that we have to show um to the field in general of of Aesthetics and neuro Aesthetics thank you Axel yeah and um I fully agreed with all that um and maybe having some artists uh invited that would be very interesting uh because um there's a lot of Applied uh active inference theory if you will and but most of the time it's applied to experts that are like us right if you if you have cogn and think in a normal way active in Prince applies to you it applies also to people who are experts in certain domains um but you know for the art it would be interesting to have like expert artists and I think in a special issue there's Robert pero is is that right yeah which what he does is super interesting and for the in a couple of second the the the short story is um that's one artist that I discussed in my Mastery thesis that I did under the supervision of sander actually so it's interesting for me this whole you know art stream because goes full circle for me uh is started with uh the aesthetic experience of well all like experience and transformative experience in productive processing right then move to Evolution biology and then you know Psychiatry and so on and so forth uh but yeah so having invited artists that would be cool and uh you know maybe pursuing those other topics that in the end all relate to aesthetic experience um you know in biology and so on and so forth you know yeah awesome I hope we can do it if someone's listened this far and they wna jump on for an art stream or suggest something that would be great so thank you fellows great to hear this and till next time bye for